

3.7.3 Evolution may lead to speciation (A-level only)

Content	Opportunities for skills development
Individuals within a population of a species may show a wide range of variation in phenotype. This is due to genetic and environmental factors. The primary source of genetic variation is mutation. Meiosis and the random fertilisation of gametes during sexual reproduction produce further genetic variation. Predation, disease and competition for the means of survival result in differential survival and reproduction, ie natural selection. Those organisms with phenotypes providing selective advantages are likely to produce more offspring and pass on their favourable alleles to the next generation. The effect of this differential reproductive success on the allele frequencies within a gene pool. The effects of stabilising, directional and disruptive selection. Evolution as a change in the allele frequencies in a population. Reproductive separation of two populations can result in the accumulation of difference in their gene pools. New species arise when these genetic differences lead to an inability of members of the populations to interbreed and produce fertile offspring. In this way, new species arise from existing species. Allopatric and sympatric speciation. The importance of genetic drift in causing changes in allele frequency in small populations. Students should be able to: • explain why individuals within a population of a species may show a wide range of variation in phenotype • explain why genetic drift is important only in small populations • explain how natural selection and isolation may result in change in the allele and phenotype frequency and lead to the formation of a new species • explain how evolutionary change over a long period of time has resulted in a great diversity of species.	MS 1.5 Students could apply their knowledge of sampling to the concept of genetic drift. PS 1.2 Students could devise an investigation to mimic the effects of random sampling on allele frequencies in a population. AT 1 Students could use computer programs to model the effects of natural selection and of genetic drift.